



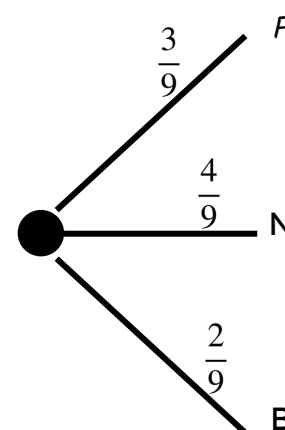
Comparing multiple representations to calculate theoretical probabilities

Task A: Outcomes and probabilities across different diagrams

list of outcomes
forward (F)
forward (F)
forward (F)
backward (B)
backward (B)
no move (N)
no move (N)
no move (N)
no move (N)

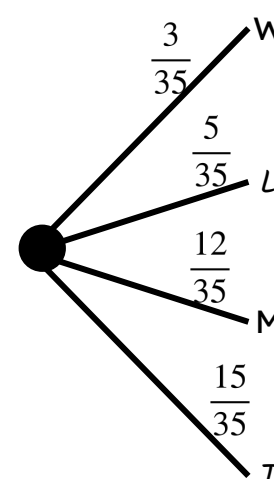
1) Andeep creates a list of equally likely outcomes from a spinner.
Use this list of outcomes to complete the frequency table and probability tree.

outcome	freq.	probability
F	3	$P(F) = \frac{3}{9}$
B	2	$P(B) = \frac{2}{9}$
N	4	$P(N) = \frac{4}{9}$
total	9	



2) Both the complete frequency table and incomplete probability tree show the same trial of randomly choosing a certain animal from a game. Complete the probability tree.

outcome	freq.	probability
lion (L)	5	$P(L) = \frac{5}{35}$
mouse (M)	12	$P(M) = \frac{12}{35}$
whale (W)	3	$P(W) = \frac{3}{35}$
turtle (T)	15	$P(T) = \frac{15}{35}$
total	35	

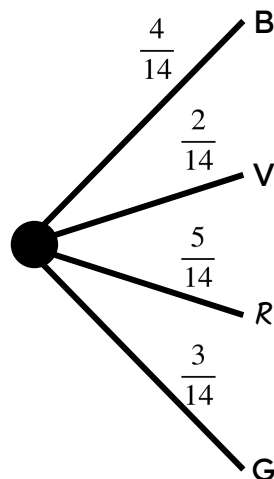


3) Both the incomplete frequency table and incomplete probability tree show the same trial of selecting a party hat of a certain colour at random.

a) Complete both diagrams.

b) Create a list showing the colour of every party hat.

outcome	freq.	probability
red (R)	5	$P(R) = \frac{5}{14}$
green (G)	3	$P(G) = \frac{3}{14}$
blue (B)	4	$P(B) = \frac{4}{14}$
violet (V)	2	$P(V) = \frac{2}{14}$
total	14	



list of outcomes

R B
R B
R B
R B
R V
G V
G
G

Task B: Showing events across different diagrams

1) This Venn diagram and outcome table show the same events.

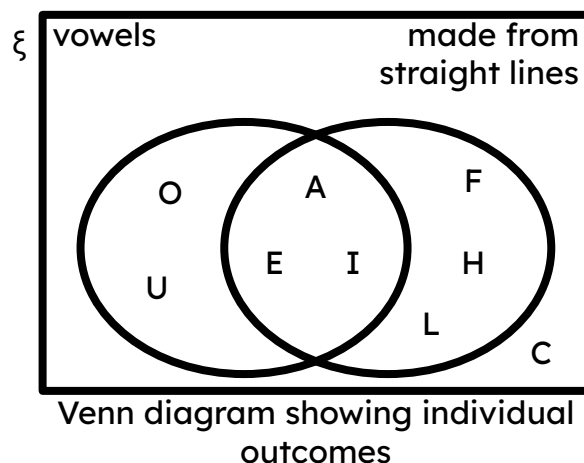
a) Use the Venn diagram to complete the outcome table.

Find these probabilities:

b) $P(\text{vowel}) = \frac{5}{9}$

c) $P(\text{not vowel}) = \frac{4}{9}$

d) $P(\text{made from straight lines}) = \frac{6}{9}$



vowels	not vowels
O A U E I	L F C H

2) A spinner has outcomes:

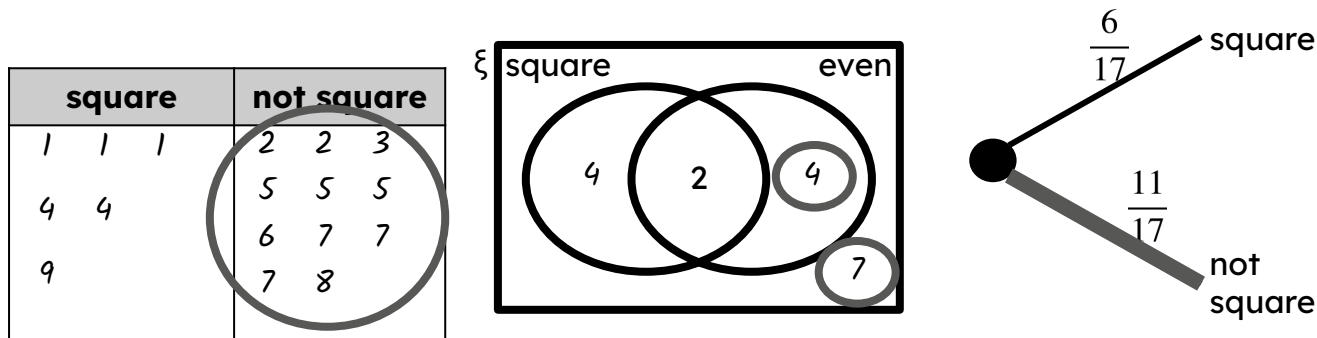
$$\xi = \{1, 1, 1, 2, 2, 3, 4, 4, 5, 5, 5, 6, 7, 7, 7, 8, 9\}$$

Trial: spinning the spinner once

Event: lands on a square number

a) Complete the outcome table, Venn diagram (showing the number of outcomes), and the probability tree for this event.

b) Circle or highlight the outcomes or branches in each diagram that are required to calculate the probability of $P(\text{not square})$.



3) A Venn diagram (showing the number of outcomes), outcome table, and probability tree all show the same event from a trial.

a) The outcome table is missing some of its outcomes!

Use the correct Venn diagram to suggest possible outcomes for the table.

One even cube number (e.g. 8), one odd cube number (e.g. 27), and one odd non-cube number should be added to the outcome table (e.g. 5).

b) Complete the probability tree.

